

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408771
Kind Code	B1
Date of Patent	September 09, 2025
Inventor(s)	Kerr; Paris

Caddy

Abstract

A caddy for holding one or a plurality of containers is disclosed. The caddy has a flexible top brace. The flexible top brace has a plurality of top contours along its length. There is a frame coupled to the flexible top brace. The frame is further coupled to a plurality of bottom contours which are located below each top contour. The bottom contour has a raised portion. The top brace and the bottom contours act together to grasp the can. This allows the can, and all of the marketing material, to be presented to the consumer with an unobstructed view.

Inventors:	Kerr; Paris (Melbourne, AU)
Applicant:	Kerr; Paris (Melbourne, AU)
Family ID:	96950310
Assignee:	DOT DESIGN GROUP PTY LTD (Melbourne, AU)
Appl. No.:	18/640057
Filed:	April 19, 2024

Publication Classification

Int. Cl.: **A47F7/28** (20060101); **A47B73/00** (20060101); **A47F7/00** (20060101); A47F5/08 (20060101)

U.S. Cl.:

CPC **A47F7/283** (20130101); **A47B73/00** (20130101); **A47F7/0028** (20130101); A47F5/08 (20130101); A47F7/28 (20130101)

Field of Classification Search

CPC: A47F (7/283); A47F (7/28); A47F (7/0028); A47F (7/0021); A47F (7/00); A47F (5/08); A47F (5/00); A47B (73/00); A47B (81/00); A47B (81/007)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1389984	12/1920	Reed	211/74	A47J 47/16
1615571	12/1926	Crecelius	206/229	A47K 1/09
1663822	12/1927	Walker	211/74	B67B 7/18
2190567	12/1939	Jung	248/108	A47F 7/283
2305965	12/1941	Henkel	211/75	A47G 23/02
D156729	12/1949	Hampe	D6/569	N/A
D164562	12/1950	Feinberg	D6/563	N/A
3224644	12/1964	Davis	222/162	B65D 83/267
3349922	12/1966	Morrison	211/113	A47G 25/746
D252429	12/1978	Morin	D6/569	N/A
4182455	12/1979	Zurawin	211/69.8	B65D 71/70
4637580	12/1986	Idstein	248/220.21	A47J 47/16
D294442	12/1987	Bordian	D6/528	N/A
4848714	12/1988	Ziaylek, Jr.	248/313	A62C 13/78
4871074	12/1988	Bryson	211/70.6	A45D 20/12
D307373	12/1989	Bearak	D7/701	N/A
4984693	12/1990	Belokin, Jr.	D6/567	A47F 5/0876
D320536	12/1990	Silbernagel	D7/619.1	N/A
D323766	12/1991	Robbins	D7/590	N/A
D329380	12/1991	Hazlett	D9/434	N/A
D331194	12/1991	Pezzoli	D9/434	N/A
5249770	12/1992	Louthan	248/230.8	A47K 1/09
5358128	12/1993	Belokin	211/75	F25D 25/02
5655673	12/1996	Weterrings	211/74	A47B 81/00
D400384	12/1997	Belokin	D6/574	N/A
D403869	12/1998	Harmon	D6/320	N/A
5913433	12/1998	Belokin	211/75	A47F 3/0486
5924579	12/1998	DuPont	211/60.1	A45D 44/02
D413473	12/1998	Belokin	D6/574	N/A
D413474	12/1998	Belokin	D6/574	N/A
6082687	12/1999	Kump	248/224.51	G09F 3/204
6102215	12/1999	Guida	211/75	B25H 3/04
D436537	12/2000	Robbins	D7/590	N/A
6279760	12/2000	Broeski	211/74	A47J 47/16
6899236	12/2004	Yang	211/75	A47B 81/00
D510015	12/2004	Hostetler	D8/373	N/A
7017759	12/2005	Friend	248/312	A47B 73/00
7028962	12/2005	Hostetler	248/205.9	B63C 11/02
D589782	12/2008	Rowell	D8/356	N/A
7556159	12/2008	Robertson	211/75	A47B 96/02
D637457	12/2010	Thompson	D7/600.1	N/A

7975858	12/2010	Schneider	211/75	A47F 5/0043
8186524	12/2011	Siahpush	211/133.4	A47F 7/283
9526356	12/2015	Pichel	N/A	A47F 5/0043
D789114	12/2016	Reed	D6/553	N/A
D790249	12/2016	Park	D6/540	N/A
D817728	12/2017	Brown	D7/701	N/A
10315321	12/2018	Frierson	N/A	A45D 44/02
D855408	12/2018	Tsai	D7/600.1	N/A
D895371	12/2019	Brown	D7/554.2	N/A
10869563	12/2019	Brenner	N/A	G09F 23/00
11033121	12/2020	Gudaitis	N/A	A47B 77/16
D953774	12/2021	Serhii	D6/552	N/A
11925280	12/2023	Brenner	N/A	G09F 7/18
2002/0158032	12/2001	Belokin	211/75	A47F 5/08
2003/0160060	12/2002	Hornblad	221/256	A47F 5/08
2003/0209505	12/2002	Belokin	211/75	A47F 7/283
2003/0222037	12/2002	Belokin	211/75	G09F 7/18
2004/0055981	12/2003	Walsh	211/75	A47F 7/28
2006/0043036	12/2005	Robertson	211/75	A47F 3/0486
2006/0054575	12/2005	Robertson	211/75	A47F 3/0486
2006/0060549	12/2005	Robertson	211/75	A47F 3/0486
2007/0152120	12/2006	Hostetler	248/206.3	B63C 11/22
2010/0148025	12/2009	Belokin	248/206.2	A47F 5/0807
2015/0250333	12/2014	Schaefer	248/205.8	A47F 5/08
2020/0093285	12/2019	Brenner	N/A	A47F 3/0486

Primary Examiner: Barnett; Devin K

Attorney, Agent or Firm: Braxton Perrone, PLLC

Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) The present invention relates to a system and method for a caddy for housing and displaying cans.

Description of Related Art

(2) Items, such as drinks, are often packaged as a can. However, the display and branding on the can is typically hidden when the can is displayed. As such, there is a need for a system to display cans without interfering with the can branding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features believed characteristic of the invention are set forth in the appended claims. The invention itself, however, as well as a preferred mode of use, further objectives and advantages thereof, will be best understood by reference to the following detailed description of illustrative embodiments when read in conjunction with the accompanying drawings, wherein:

- (2) FIG. 1 is a perspective view of the caddy in one embodiment;
- (3) FIG. 2 is a rear perspective view of the caddy in one embodiment;
- (4) FIG. 3 is a top view of the caddy in one embodiment;
- (5) FIG. 4 is a left side view of the caddy in one embodiment;
- (6) FIG. 5 is a front view of the caddy in one embodiment;
- (7) FIG. 6 is a right side view of the caddy in one embodiment;
- (8) FIG. 7 is a bottom perspective of a can in one embodiment;
- (9) FIG. 8 is a top front perspective view of a display in one embodiment;
- (10) FIG. 9 is a top rear perspective view of a display in one embodiment;
- (11) FIG. 10 is a top view of a display in one embodiment;
- (12) FIG. 11 is a left side view of a display in one embodiment;
- (13) FIG. 12 is a front side view of a display in one embodiment;
- (14) FIG. 13 is a right side view of a display in one embodiment;
- (15) FIG. 14 is a view of a display being coupled to a caddy in one embodiment;
- (16) FIG. 15 is an assembled display and caddy in one embodiment;
- (17) FIG. 16 is an exploded view of the suction device in one embodiment;
- (18) FIG. 17 is a side perspective view of a can in a caddy in one embodiment;
- (19) FIG. 18 is a bottom perspective view of a can sitting on a bottom contour in one embodiment.

DETAILED DESCRIPTION

(20) Several embodiments of Applicant's invention will now be described with reference to the drawings. Unless otherwise noted, like elements will be identified by identical numbers throughout all figures. The invention illustratively disclosed herein suitably may be practiced in the absence of any element which is not specifically disclosed herein.

(21) FIG. 1 is a perspective view of the caddy in one embodiment. A caddy, as used herein, refers to an item which is used to house, display, and hold one or more objects. In some embodiments, the caddy is used to house and display one or more cans.

(22) The caddy **100** can comprise virtually any material. It can comprise plastic, rubber, metals, and combinations thereof. In one embodiment, and as described in more detail below, some components are flexible whereas other components are rigid. In one embodiment the material for the caddy **100** comprises polycarbonate. Polycarbonate, in some embodiments, has a flexural yield strength of 36-103 MPa. When used in the caddy **100** it is able to flex ± 35 mm from neutral without breaking.

(23) As shown, the caddy **100** has a top brace **101** and a frame **103**. The frame **103** has a plurality of bottom contours **102**. A frame **103** extends for the length of the caddy **100** and provides structure. In one embodiment the frame **103** is rigid. In one embodiment the frame **103** has some flex to compensate for the weight of the cans. In one embodiment the frame **103** curves upward by 3 degrees in the center. This compensates for the weight of the inserted cans. When the cans are inserted, the frame **103** flattens to be flat.

(24) The top brace **101**, as shown, and in one embodiment, has a plurality of curves, or top contours **101**, which are designed to mimic the outer diameter of the can. The caddy **100** maintains contact with the surface by use of a pressure mechanism and geometry. The caddy **100** secures the cans or bottles by use of an inward concaved top brace **101** which extends along the length of the caddy **100** between suction cup attachments. As will be shown, the top brace **101** inverts to secure the can or bottle securely between the top brace **101** and glass surface with minimal material usage. Further, as can be appreciated, if viewed by a consumer from the glass side, because the top brace **101** is on the rear side, behind the can or bottle, the branding on the can or bottle is easily viewable by the consumer with minimal or no obstruction. As will be described below, brands spend considerable time and resources designing a packaging which will entice the consumer. Covering up all or portion of the branding/label frustrates this purpose. Many such caddies exist which entirely or partially obscure the branding/label. This is undesirable and voids much of the market

research which goes into designing the label. As shown, the caddy **100** allows for housing of the can/bottle with minimal or no obstruction of the package when viewed from the glass side.

(25) As shown the top brace **101** extends the length of the caddy **100**. The top brace **101** is on the top side of the caddy **100**. The top brace **101** engages with the top or side portion of the can or bottle. While the description references a can, this is for illustrative purposes only and should not be deemed limiting. The caddy **100** can house and display virtually any type of containers including bottles, cans, etc.

(26) In one embodiment, the radius edge on the top brace **101** is small to avoid can branding distortion yet smooth enough not to pierce the can when inserted or pulled out.

(27) Turning briefly to FIG. 7, FIG. 7 is a bottom perspective of a can in one embodiment. As can be seen, the can has a can side **112** which extends upwardly from the bottom of the can to the top of the can. The can side **112** comprises an outer curvature. In one embodiment, the top brace **101** at least one top contour **118** which comprises a curvature which matches the curvature of the can side **112**. Thus, if a larger can is utilized, a top contour **118** with a corresponding curvature which can accommodate the larger diameter is utilized.

(28) As can be seen, the bottom of the can comprises a lip **113**. In the embodiment depicted the lip **113** comprises an enclosed shape—in this case, an enclosed circular lip portion. When upright, the can will rest upon the lip **113**. Turning back to FIG. 1, FIG. 1 shows a bottom contour **102**. The bottom contour **102** engages with the bottom of the can or bottle. The bottom contour **102** can either engage outside of the lip **113** or inside the lip **113**. As shown the bottom contour **102** has a smaller effective diameter than the top contour **118**. In this embodiment, the top brace **101** interacts with the outside of the can whereas the bottom contour **102** acts on the inside of the lip **113**. This interaction helps keep the can in its desired location.

(29) In one embodiment, and as depicted, a portion of the bottom contour **102** raises upward to allow it to fit within the enclosed shape defined by the lip **113** of the can. Specifically, the bottom contour has a raised portion **114** which extends upward. The can will rest upon the raised portion **114**. In some embodiments, the raised portion **114** sits adjacent the lip **113**. This helps secure the can in its desired location. Thus, the can rests atop the bottom contour **102**, and specifically, the raised portion **114**. In one embodiment the raised portion **114** is raised about 3 mm above the bottom contour **102**. However, in other embodiments it can be extended to about 5 mm to accommodate cans of different dimensions.

(30) Turning briefly to FIG. 18, FIG. 18 is a bottom perspective view of a can sitting on a bottom contour in one embodiment. As can be seen, the bottom contour **102** has a raised portion **114** which raises up and interacts with the can. The lip **113** of the can is adjacent to the raised portion **114**. In one embodiment, the raised portion **114** comprises a shape which mimics the enclosed shape of the lip **113**. Since the lip **113** creates a circular shape, the raised portion **114** also forms a partial circle so as to reside within, and adjacent to, the enclosed shape created by the lip **113**.

(31) As shown, the bottom contour **102** comprises a bottom contour void **117**. This is a void **117** in the bottom contour **102**. This provides a place for the user's finger to grip the can and lift upwards to retrieve a can. As can be seen, due to the void, there is sufficient space for a finger to fit within which can dislodge the can from resting upon the raised portion **114**.

(32) Turning to FIG. 17, FIG. 17 is a side perspective view of a can in a caddy. As can be seen, the top brace **101**, and accordingly, the top contour **118**, couples to the can side **112**. As noted above, the bottom contour **112** engages with the bottom of the can.

(33) Turning to FIG. 2, FIG. 2 is a rear perspective view of the caddy in one embodiment. This is what from the side of a transparent structure that the consumer would see. This is referred to as the transparent side. The transparent structure can comprise virtually any structure which is at least partially transparent. This can include doors, windows, etc. Often the transparent structure is a transparent portion of a door or window within a refrigerator, freezer, or the like. Thus, the consumer can view the contents of the refrigerator, for example. The transparent structure can

comprise virtually any material, including but not limited to, glass, plastics, etc.

(34) As can be appreciated, if the caddy **100** is pressed up against glass, for example, such that the suction cups (not shown in this figure) are attached to the glass structure, then the consumer experiences an unobstructed view of the can/bottle. Put differently, no part of the caddy is between the can and the glass structure. Thus, no part of the caddy will obstruct the view of the packaging/label when viewed from the transparent side (what a consumer sees when looking into a closed glass door).

(35) Turning to FIG. 3, FIG. 3 is a top view of the caddy **100** in one embodiment. As seen from this view, the top contour **118** often has a greater diameter than the bottom contour **102**.

(36) FIG. 3 also illustrates that in some embodiments, and as depicted, the top brace **101** has flexibility or pliability. Put differently, the top brace **101** is not rigid but is instead flexible. This flexibility applies an internal pressure against the can. In one embodiment the top brace **101** applied a force of up to 1.5 Kgs on the cans closest to the suction cups and 0.5 Kgs for the cups further from the cups. As can be seen, the top brace **101** has five distinct top contours **118**. Each top contour **118** can accommodate a separate can. The first top contour **118**, on the extreme left, has a greater distance from the contour to the frame **103**. However, the second can contour has a decreased distance from the top contour **118** to the frame **103**. This is because the top brace **101** has flex. As shown, when there are no cans, the top brace **101** is in a relaxed state. However, by applying an outward force, a new can could be inserted. The flex of the top brace **101** will now apply a force against the can. This helps assure the can stays in the desired location.

(37) FIG. 4 is a left side view of the caddy **100** in one embodiment. FIG. 4 also shows the relative sides of the top and bottom contours. FIG. 4 also shows a feature **115** at the bottom of the caddy **100**. The feature **115**, in some embodiments, is a small feature which extends outward from the bottom of the caddy **100**. As shown, the feature **115** has a reduced diameter portion **116** which has a smaller diameter compared to the remaining portion of the feature **115**. As will be described in more detail below, this feature **115** allows for the coupling of the display **106**.

(38) FIG. 5 is a front view of the caddy **100** in one embodiment. As can be seen, the suction cups are at either end of the caddy **100**. In one embodiment the suction cup has space for marketing applications. For example, point of sale information can be displayed on the suction cups.

(39) FIG. 6 is a right-side view of the caddy in one embodiment.

(40) Turning to FIG. 8, FIG. 8 is a top front perspective view of an optional display **106** in one embodiment. Similarly, FIG. 9 is a top rear perspective view of a display in one embodiment.

(41) The display **106** can display any desired sales or POS information. This can include price, brand, sales information, etc. In one embodiment, the display **106** couples to the caddy **100** in such a way that the branding or label is not obstructed by the display **106**. Instead, the display **106** is shown separately from the branding/label.

(42) The display **106** can couple to the caddy **100** via any method or device known in the art. In one embodiment the display **106** comprises a coupler **107**. The coupler **107** couples to a corresponding feature **115** on the caddy **100**. In one embodiment the coupler **107** has a friction fit which surrounds and receives a feature **115** on the caddy **100**. This allows the display **106** to couple and de-couple without any tools. Instead, the coupler **107** simply slides onto and attaches to the caddy **100**.

(43) Turning to FIG. 10, FIG. 10 is a top view of a display in one embodiment. FIG. 11 is a left side view of a display in one embodiment. FIG. 11 shows the open coupler **107** which can receive a feature **115** from the caddy **100**. This feature allows the display **106** to be easily added or removed from the caddy **100**.

(44) FIG. 12 is a front side view of a display in one embodiment. FIG. 13 is a right-side view of a display in one embodiment.

(45) FIG. 14 is a view of a display **106** being coupled to a caddy **100** in one embodiment. As can be seen, the display **106** simply slides over a lower feature **115** on the caddy **100**. The coupler **107**

receives this feature **115**. The coupler **107** rests at the reduced diameter **116** on the feature **115**.

(46) FIG. **15** is an assembled display and caddy in one embodiment showing the display **106** coupled to the caddy **100**.

(47) Turning to FIG. **16**, FIG. **16** is an exploded view of the suction device in one embodiment. While one embodiment of a suction device is discussed, this is for illustrative purposes only and should not be deemed limiting. Virtually any type of suction device can be utilized.

(48) As shown, the caddy has a void structure **105**. This provides a unique structure which can be coupled to the suction structure **110**. This ensures the suction is properly oriented for coupling. The suction structure **110** fits within the void structure **105**. In one embodiment the two pieces are friction fitted. The system, in one embodiment, further comprises a stem **109** which is coupled to a control **108**. Manipulation of the control **108**, such as by rotating for example, moves the stem **109**. The stem **109** likewise moves the suction cup **111**. Thus, generating or releasing the suction force applied by the suction cup **111** can be controlled by the control **108**.

(49) While the caddy **100** is shown as being horizontal, this is for illustrative purposes only and should not be deemed limiting. The design can be used with vertical alignment of cans as well.

(50) The suction shown, in one embodiment, has a weight limit of about 27 kg per suction.

(51) The system has many advantages. First, as noted, in some embodiments, no portion of the caddy extends between the can and the transparent structure, such as glass. This means that when the user views the can from the glass side, the consumer has a completely unobstructed view. The entirety of the can or bottle labeling can be viewed by the consumer. This is ideal as the manufacturer knows what they want to present to the consumer, and a caddy which interferes with the delivery of the marketing material, is undesirable.

(52) Additionally, this provides an unexpected aesthetic benefit. Rather than having a portion of the caddy visible, the user sees a clean view of the can presented adjacent to the transparent structure.

(53) Another advantage is the ability to securely nest the can or bottle in the desired location. The caddy uses a flexible top brace which applies a force against the can to press the can against the transparent structure and keep it secure. The caddy also uses a bottom contour to grasp and secure the can/bottle into the desired location. The use of the flexible top brace along with the bottom contour securely and safely holds the item securely.

(54) Another advantage is that the inverted arc presses cans, for example, against the glass with minimal contact and holds the can firm with vibration when the door is closed or opened.

(55) Another advantage is the decreased weight to capacity achieved by the caddy. The caddy reduces raw materials usage compared to a prior caddy. In one embodiment the caddy **100** uses between about 50-70% less plastic than industry suction caddies. In one embodiment the caddy **100** uses about 68% less plastic than industry suction caddies. In previous caddies the weight was about 0.97 pounds, whereas the instant caddy is about 0.514 pounds.

(56) Another advantage, in some embodiments, is that caddies can nest into each other to reduce space in freight meaning less air is freighted in each bod.

(57) Another advantage is the ability to place the caddy adjacent any transparent structure. This increases visibility and flexibility. The caddy offers an additional opportunity to display containers for sale.

(58) While the invention has been particularly shown and described with reference to a preferred embodiment, it will be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the invention.

Claims

1. A caddy for holding a plurality of containers, said caddy comprising: a single flexible top brace having a plurality of top contours along its length; a frame coupled to said flexible top brace; wherein said frame is further coupled to a plurality of bottom contours; wherein each bottom

contour is located below said top contours; wherein each bottom contour comprises a raised portion which extends upward and upon which one of said plurality of containers will rest, and wherein said top contours comprise a greater diameter than said bottom contours; suction cups on each side of said caddy, and wherein said flexible top brace extends between said suction cups.

2. The caddy of claim 1 wherein said frame is rigid.
3. The caddy of claim 1 wherein each of said bottom contours comprises a semi-circular shape.
4. The caddy of claim 1 wherein each of said bottom contours comprises a bottom contour void.
5. The caddy of claim 1 further comprising a void structure on each side of said caddy.
6. The caddy of claim 1 further comprising suction cups on each side of said caddy.
7. The caddy of claim 1 further comprising at least one can located within said flexible top brace, wherein said at least one can is supported by said bottom contours.
8. The caddy of claim 7 wherein said at least one can comprises a can lip, and wherein said can lip forms an enclosed shape, and wherein said raised portion of one of said bottom contours fits within said enclosed shape.
9. The caddy of claim 7 wherein said caddy is coupled to a transparent structure, and wherein there is no obstruction of view of the at least one can from a transparent side.
10. The caddy of claim 9 wherein no part of said caddy is between said at least one can and said transparent structure.
11. The caddy of claim 7 wherein said flexible top brace applies a force against said at least one can.
12. The caddy of claim 1 wherein said plurality of contours comprises five top contours.
13. The caddy of claim 1 wherein said caddy comprises a feature along its bottom.
14. The caddy of claim 13 wherein said feature comprises a reduced diameter portion.
15. The caddy of claim 13 further comprising a display coupled to said feature.
